2.2 Speed and Velocity

Section Learning Objectives

By the end of this section, you will be able to do the following:

- Calculate the average speed of an object
- Relate displacement and average velocity

Teacher Support

Teacher Support

The learning objectives in this section will help your students master the following standards:

- (4) Science concepts. The student knows and applies the laws governing motion in a variety of situations. The student is expected to:
    - (B) describe and analyze motion in one dimension using equations with the concepts of distance, displacement, speed, average velocity, instantaneous velocity, and acceleration.

In addition, the High School Physics Laboratory Manual addresses content in this section in the lab titled: Position and Speed of an Object, as well as the following standards:

- (4) Science concepts. The student knows and applies the laws governing motion in a variety of situations. The student is expected to:
    - (B) describe and analyze motion in one dimension using equations with the concepts of distance, displacement, speed, average velocity, instantaneous velocity, and acceleration.

Section Key Terms

average speed	average velocity	instantaneous speed
instantaneous velocity	speed	velocity

Teacher Support

Teacher Support

In this section, students will apply what they have learned about distance and displacement to the concepts of speed and velocity.

[BL][OL] Before students read the section, ask them to give examples of ways they have heard the word speed used. Then ask them if they have heard the word velocity used. Explain that these words are often used interchangeably in everyday life, but their scientific definitions are different. Tell students that they will learn about these differences as they read the section.

[AL] Explain to students that velocity, like displacement, is a vector quantity. Ask them to speculate about ways that speed is different from velocity. After they share their ideas, follow up with questions that deepen their thought process, such as: Why do you think that? What is an example? How might apply these terms to motion that you see every day?

Speed

There is more to motion than distance and displacement. Questions such as, “How long does a foot race take?” and “What was the runner’s speed?” cannot be answered without an understanding of other concepts. In this section we will look at time, speed, and velocity to expand our understanding of motion.

A description of how fast or slow an object moves is its speed. Speed is the rate at which an object changes its location. Like distance, speed is a scalar because it has a magnitude but not a direction. Because speed is a rate, it depends on the time interval of motion. You can calculate the elapsed time or the change in time, Δt, of motion as the difference between the ending time and the beginning time

$$\Delta t = t_f - t_0 \ .$$

The SI unit of time is the second (s), and the SI unit of speed is meters per second (m/s), but sometimes kilometers per hour (km/h), miles per hour (mph) or other units of speed are used.

When you describe an object's speed, you often describe the average over a time period. Average speed, v_{avg}, is the distance traveled divided by the time during which the motion occurs.

$$v_{\text{avg}} = \frac{\text{distance}}{\text{time}}$$

You can, of course, rearrange the equation to solve for either distance or time

$$\text{time} = \frac{\text{distance}}{v_{\text{avg}}}.$$

$$\text{distance} = v_{\text{avg}} \times \text{time}$$

Suppose, for example, a car travels 150 kilometers in 3.2 hours. Its average speed for the trip is

$$\begin{aligned} v_{\text{avg}} &= \frac{\text{distance}}{\text{time}} \\ &= \frac{150 \text{ km}}{3.2 \text{ h}} \\ &= 47 \text{ km/h}. \end{aligned}$$

A car's speed would likely increase and decrease many times over a 3.2 hour trip. Its speed at a specific instant in time, however, is its instantaneous speed. A car's speedometer describes its instantaneous speed.

Teacher Support

Teacher Support

[OL][AL] Caution students that average speed is not always the average of an object's initial and final speeds. For example, suppose a car travels a distance of 100 km. The first 50 km it travels 30 km/h and the second 50 km it travels at 60 km/h. Its average speed would be distance /(time interval) = (100 km)/[(50 km)/(30 km/h) + (50 km)/(60 km/h)] = 40 km/h. If the car had spent equal times at 30 km and 60 km rather than equal distances at these speeds, its average speed would have been 45 km/h.

[BL][OL] Caution students that the terms speed, average speed, and instantaneous speed are all often referred to simply as speed in everyday language. Emphasize the importance in science to use correct terminology to avoid confusion and to properly communicate ideas.

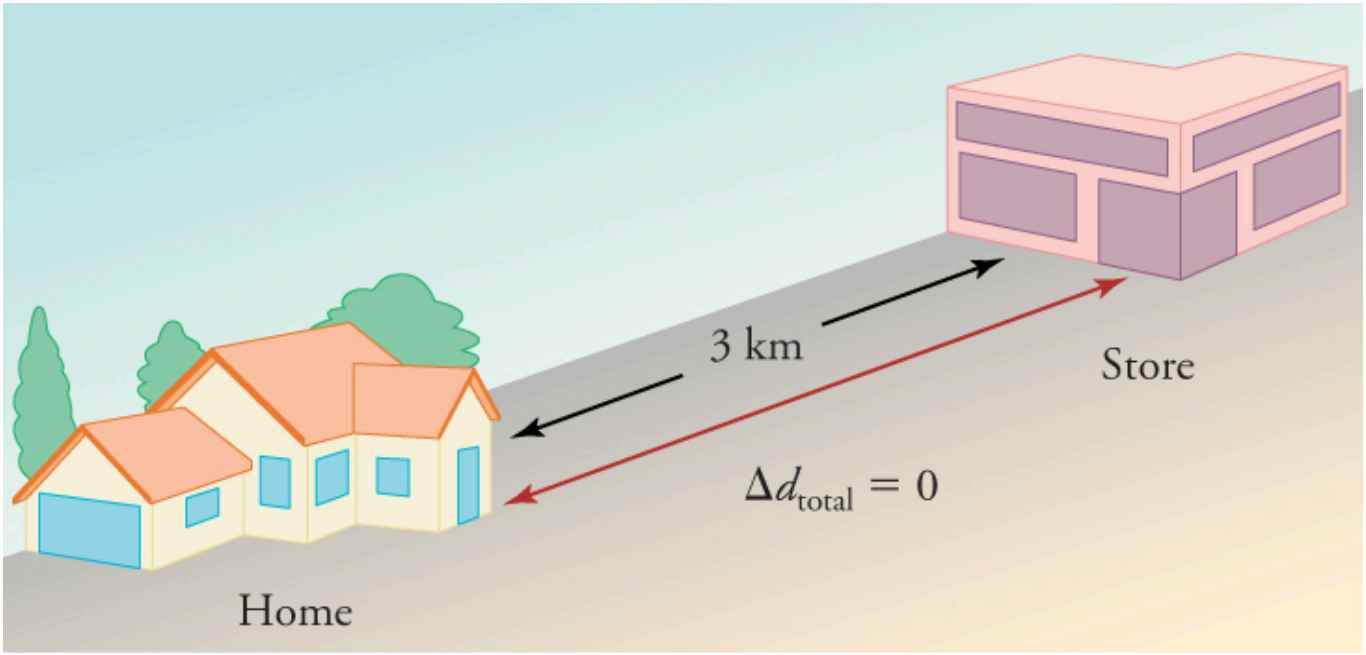


Figure 2.8 During a 30-minute round trip to the store, the total distance traveled is 6 km. The average speed is 12 km/h. The displacement for the round trip is zero, because there was no net change in position.

Worked Example

Calculating Average Speed

A marble rolls 5.2 m in 1.8 s. What was the marble's average speed?

Strategy

We know the distance the marble travels, 5.2 m, and the time interval, 1.8 s. We can use these values in the average speed equation.

Solution

$$v_{\text{avg}} = \frac{\text{distance}}{\text{time}} = \frac{5.2\ \text{m}}{1.8\ \text{s}} = 2.9\ \text{m/s}$$

Discussion

Average speed is a scalar, so we do not include direction in the answer. We can check the reasonableness of the answer by estimating: 5 meters divided by 2 seconds is 2.5 m/s. Since 2.5 m/s is close to 2.9 m/s, the answer is reasonable. This is about the speed of a brisk walk, so it also makes sense.

Practice Problems

9.
A pitcher throws a baseball from the pitcher’s mound to home plate in 0.46 s. The distance is 18.4 m. What was the average speed of the baseball?

a. 40 m/s
b. - 40 m/s
c. 0.03 m/s
d. 8.5 m/s

10 .
Cassie walked to her friend's house with an average speed of 1.40 m/s. The distance between the houses is 205 m. How long did the trip take her?

a. 146 s
b. 0.01 s
c. 2.50 min
d. 287 s

Velocity

The vector version of speed is velocity. Velocity describes the speed and direction of an object. As with speed, it is useful to describe either the average velocity over a time period or the velocity at a specific moment. Average velocity is displacement divided by the time over which the displacement occurs.

$$\mathbf{v}_{\text{avg}} = \frac{\text{displacement}}{\text{time}} = \frac{\Delta \mathbf{d}}{\Delta t} = \frac{\mathbf{d}_f - \mathbf{d}_0}{t_f - t_0}$$

Velocity, like speed, has SI units of meters per second (m/s), but because it is a vector, you must also include a direction. Furthermore, the variable **v** for velocity is bold because it is a vector, which is in contrast to the variable *v* for speed which is italicized because it is a scalar quantity.

Tips For Success

It is important to keep in mind that the average speed is not the same thing as the average velocity without its direction. Like we saw with displacement and distance in the last section, changes in direction over a time interval have a bigger effect on speed and velocity.

Suppose a passenger moved toward the back of a plane with an average velocity of –4 m/s. We cannot tell from the average velocity whether the passenger stopped momentarily or backed up before he got to the back of the plane. To get more details, we must consider smaller segments of the trip over smaller time intervals such as those shown in Figure 2.9. If you consider infinitesimally small intervals, you can define instantaneous velocity, which is the velocity at a specific instant in time. Instantaneous velocity and average velocity are the same if the velocity is constant.

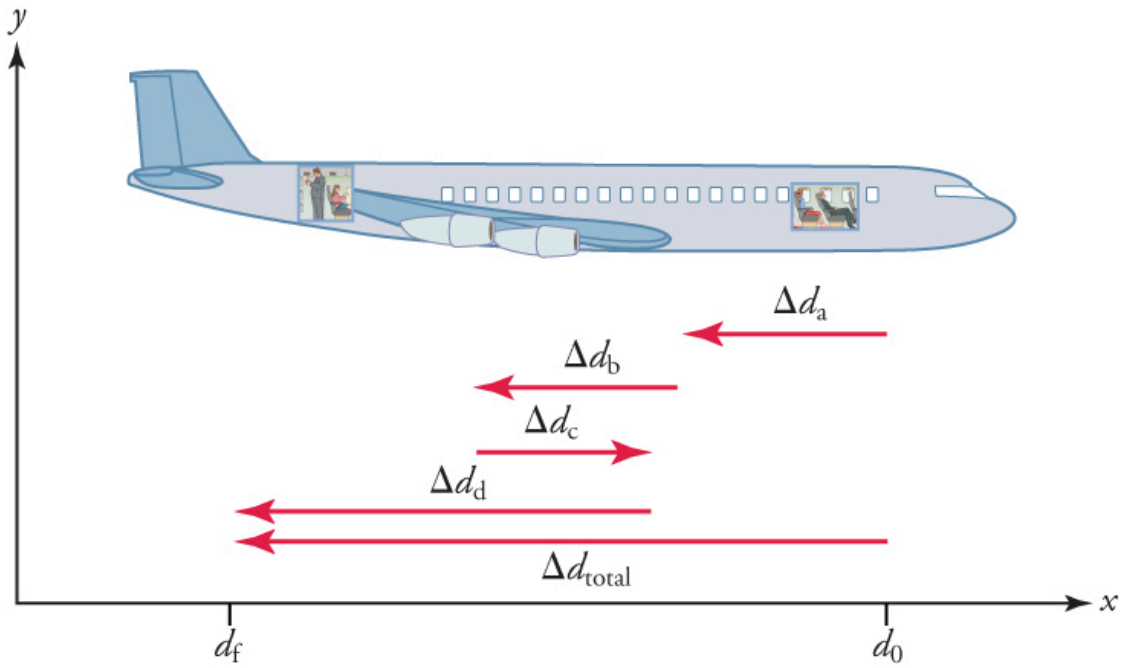


Figure 2.9 The diagram shows a more detailed record of an airplane passenger heading toward the back of the plane, showing smaller segments of his trip.

Earlier, you have read that distance traveled can be different than the magnitude of displacement. In the same way, speed can be different than the magnitude of velocity. For example, you drive to a store and return home in half an hour. If your car's odometer shows the total distance traveled was 6 km, then your average speed was 12 km/h. Your average velocity, however, was zero because your displacement for the round trip is zero.

Watch Physics

Calculating Average Velocity or Speed

This video reviews vectors and scalars and describes how to calculate average velocity and average speed when you know displacement and change in time. The video also reviews how to convert km/h to m/s.

Average velocity for constant acceleration | One-dimensional motion | Physics | Khan Academy

Which of the following fully describes a vector and a scalar quantity and correctly provides an example of each?

a. A scalar quantity is fully described by its magnitude, while a vector needs both magnitude and direction to fully describe it. Displacement is an example of a scalar quantity and time is an example of a vector quantity.
b. A scalar quantity is fully described by its magnitude, while a vector needs both magnitude and direction to fully describe it. Time is an example of a scalar quantity and displacement is an example of a vector quantity.
c. A scalar quantity is fully described by its magnitude and direction, while a vector needs only magnitude to fully describe it. Displacement is an example of a scalar quantity and time is an example of a vector quantity.
d. A scalar quantity is fully described by its magnitude and direction, while a vector needs only magnitude to fully describe it. Time is an example of a scalar quantity and displacement is an example of a vector quantity.

Teacher Support

Teacher Support

This video does a good job of reinforcing the difference between vectors and scalars. The student is introduced to the idea of using ‘s’ to denote displacement, which you may or may not wish to encourage. Before students watch the video, point out that the instructor uses $\vec{s}$ for displacement instead of d, as used in this text. Explain the use of small arrows over variables is a common way to denote vectors in higher-level physics courses. Caution students that the customary abbreviations for hour and seconds are not used in this video. Remind students that in their own work they should use the abbreviations h for hour and s for seconds.

Worked Example

Calculating Average Velocity

A student has a displacement of 304 m north in 180 s. What was the student's average velocity?

Strategy

We know that the displacement is 304 m north and the time is 180 s. We can use the formula for average velocity to solve the problem.

Solution

$$\mathbf{v}_{\text{avg}} = \frac{\Delta \mathbf{d}}{\Delta t} = \frac{304 \text{ m}}{180 \text{ s}} = 1.7 \text{ m/s north} \qquad 2.1$$

Discussion

Since average velocity is a vector quantity, you must include direction as well as magnitude in the answer. Notice, however, that the direction can be omitted until the end to avoid cluttering the problem. Pay attention to the significant figures in the problem. The distance 304 m has three significant figures, but the time interval 180 s has only two, so the quotient should have only two significant figures.

Tips For Success

Note the way scalars and vectors are represented. In this book d represents distance and displacement. Similarly, v represents speed, and v represents velocity. A variable that is not bold indicates a scalar quantity, and a bold variable indicates a vector quantity. Vectors are sometimes represented by small arrows above the variable.

Teacher Support

Teacher Support

Use this problem to emphasize the importance of using the correct number of significant figures in calculations. Some students have a tendency to include many digits in their final calculations. They incorrectly believe they are improving the accuracy of their answer by writing many of the digits shown on the calculator. Point out that doing this introduces errors into the calculations. In more complicated calculations, these errors can propagate and cause the final answer to be wrong. Instead, remind students to always carry one or two extra digits in intermediate calculations and to round the final answer to the correct number of significant figures.

Worked Example

Solving for Displacement when Average Velocity and Time are Known

Layla jogs with an average velocity of 2.4 m/s east. What is her displacement after 46 seconds?

Strategy

We know that Layla's average velocity is 2.4 m/s east, and the time interval is 46 seconds. We can rearrange the average velocity formula to solve for the displacement.

Solution

$$\begin{aligned} \mathbf{v}_{\text{avg}} &= \frac{\Delta \mathbf{d}}{\Delta t} \\ \Delta \mathbf{d} &= \mathbf{v}_{avg}\,\Delta t \\ &= (2.4\ \text{m/s})(46\ \text{s}) \\ &= 1.1 \times 10^{2}\ \text{m east} \end{aligned} \qquad 2.2$$

Discussion

The answer is about 110 m east, which is a reasonable displacement for slightly less than a minute of jogging. A calculator shows the answer as 110.4 m. We chose to write the answer using scientific notation because we wanted to make it clear that we only used two significant figures.

Tips For Success

Dimensional analysis is a good way to determine whether you solved a problem correctly. Write the calculation using only units to be sure they match on opposite sides of the equal mark. In the worked example, you have
m = (m/s)(s). Since seconds is in the denominator for the average velocity and in the numerator for the time, the unit cancels out leaving only m and, of course, m = m.

Worked Example

Solving for Time when Displacement and Average Velocity are Known

Phillip walks along a straight path from his house to his school. How long will it take him to get to school if he walks 428 m west with an average velocity of 1.7 m/s west?

Strategy

We know that Phillip's displacement is 428 m west, and his average velocity is 1.7 m/s west. We can calculate the time required for the trip by rearranging the average velocity equation.

Solution

$$\begin{aligned} \mathbf{v}_{\text{avg}} &= \frac{\Delta \mathbf{d}}{\Delta t} \\ \Delta t &= \frac{\Delta \mathbf{d}}{\mathbf{v}_{\text{avg}}} \\ &= \frac{428\ \text{m}}{1.7\ \text{m/s}} \\ &= 2.5 \times 10^{2}\ \text{s} \end{aligned} \qquad 2.3$$

Discussion

Here again we had to use scientific notation because the answer could only have two significant figures. Since time is a scalar, the answer includes only a magnitude and not a direction.

Practice Problems

11 .
A trucker drives along a straight highway for 0.25 h with a displacement of 16 km south. What is the trucker’s average velocity?

a. 4 km/h north
b. 4 km/h south
c. 64 km/h north
d. 64 km/h south

12.
A bird flies with an average velocity of 7.5 m/s east from one branch to another in 2.4 s. It then pauses before flying with an average velocity of 6.8 m/s east for 3.5 s to another branch. What is the bird's total displacement from its starting point?

a. 42 m west
b. 6 m west
c. 6 m east
d. 42 m east

Virtual Physics

The Walking Man

In this simulation you will put your cursor on the man and move him first in one direction and then in the opposite direction. Keep the *Introduction* tab active. You can use the *Charts* tab after you learn about graphing motion later in this chapter. Carefully watch the sign of the numbers in the position and velocity boxes. Ignore the acceleration box for now. See if you can make the man's position positive while the velocity is negative. Then see if you can do the opposite.

Grasp Check

Which situation correctly describes when the moving man's position was negative but his velocity was positive?

a. Man moving toward 0 from left of 0
b. Man moving toward 0 from right of 0
c. Man moving away from 0 from left of 0
d. Man moving away from 0 from right of 0

Teacher Support

Teacher Support

This is a powerful interactive animation, and it can be used for many lessons. At this point it can be used to show that displacement can be either positive or negative. It can also show that when displacement is negative, velocity can be either positive or negative. Later it can be used to show that velocity and acceleration can have different signs. It is strongly suggested that you keep students on the *Introduction* tab. The *Charts* tab can be used after students learn about graphing motion later in this chapter.

Check Your Understanding

13 .
Two runners traveling along the same straight path start and end their run at the same time. At the halfway mark, they have different instantaneous velocities. Is it possible for their average velocities for the entire trip to be the same?

a. Yes, because average velocity depends on the net or total displacement.
b. Yes, because average velocity depends on the total distance traveled.
c. No, because the velocities of both runners must remain exactly the same throughout the journey.
d. No, because the instantaneous velocities of the runners must remain the same at the midpoint but can vary at other points.

14.
If you divide the total distance traveled on a car trip (as determined by the odometer) by the time for the trip, are you calculating the average speed or the magnitude of the average velocity, and under what circumstances are these two quantities the same?

a. Average speed. Both are the same when the car is traveling at a constant speed and changing direction.
b. Average speed. Both are the same when the speed is constant and the car does not change its direction.
c. Magnitude of average velocity. Both are same when the car is traveling at a constant speed.
d. Magnitude of average velocity. Both are same when the car does not change its direction.

15 .
Is it possible for average velocity to be negative?

a. Yes, if net displacement is negative.
b. Yes, if the object's direction changes during motion.
c. No, because average velocity describes only the magnitude and not the direction of motion.
d. No, because average velocity only describes the magnitude in the positive direction of motion.

Teacher Support

Teacher Support

Use the *Check Your Understanding* questions to assess students' achievement of the sections learning objectives. If students are struggling with a specific objective, the *Check Your Understanding* will help identify which and direct students to the relevant content. Assessment items in TUTOR will allow you to reassess.

"2.2 Speed and Velocity" from Physics by Paul Peter Urone, Roger Hinrichs, OpenStax.

Licensed under Creative Commons Attribution License (CC BY 4.0). http://creativecommons.org/licenses/by/4.0/

Original page: https://openstax.org/books/physics/pages/2-2-speed-and-velocity

Access for free at openstax.org.

Changes: reformatted as a printable page for evaluating ReadableSTEM; image alt text left out of this print; no wording edited.

ReadableSTEM is not affiliated with or endorsed by OpenStax.